

Memory-Centric Agentic Inference — cycles 16-18

2026-05-11

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Memory-Centric Agentic Inference — cycles 16-18

Abstract

Cycles 16-18 moved the memory-centric architecture package from synthesis and measurement design into two executable falsification layers: an energy/economics and CXL-contention harness, and a trace-v3 security-enforcement replay. The central architecture thesis remains conditional: future AI infrastructure should expose and manage memory objects only when retained state value survives coordination cost, validation overhead, recovery cost, expected security loss, durable-tail cost, and now measured or bounded energy, dollar, and contention costs.

The main result of cycle 16 was **M-ENERGY-1**, a synthetic sensitivity harness for two deferred constants: **DC-001**, per-tier energy and dollar cost per byte moved or retained, and **DC-002**, CXL or pooled-memory latency under contention. It produced 288 scenario rows, 96 architecture-sensitivity rows, 48 CXL threshold rows, 7 claim-update rows, 4 measurement-requirement rows, and three figures. The harness does not claim measured energy savings; it defines what would support or falsify such claims.

The main result of cycle 17 was **M-SECOPS-1**, a trace-v3 security telemetry and enforcement replay. It extends the prior synthetic trace with tenant scope, cache salt, actor identity, replay authorization, verifier evidence hashes, retention state, pointer validity, validation gates, and validation timing. The replay produced 503 trace-v3 rows, 268 enforcement decisions, 13 fixtures, 54 field-ablation rows, and 6 architecture-update rows. After audit repair, invalid fixtures execute through the same decision path as replay rows, and unsafe reuse receives zero safe reuse credit.

Cycle 18, as reflected in the supplied session IDs and current workspace state, did not add a separate named artifact beyond the already validated **M-ENERGY-1** and **M-SECOPS-1** package. It serves here as the reporting and consolidation boundary for cycles 16-18. A source-access gap remains: no callable `search_sessions` or `list_session_catalog` tool was available, so the report uses the supplied session IDs as traceability anchors and relies on workspace artifacts, ledger records, validation outputs, and the supplied audit report for technical content.

Introduction

The project investigates whether agentic large-language-model inference should be designed around memory movement, placement, reuse, compression, and lifetime management rather than arithmetic throughput alone. Earlier cycles built a taxonomy, lifetime model, heterogeneous memory cost model, simulator, scheduling analysis, architecture proposal, trace schema, queueing model, compression model, runtime prototype, calibration map, security/provenance model, synthesis matrix, and measurement plans.

At the start of this reporting window, the package had already converged on three architecture options:

- **Option A: conventional request/model/KV serving.** This is the control baseline. It treats most work as request-local and does not expose durable cross-request or trajectory-level memory state as first-class retained value.
- **Option B: memory-object-aware runtime.** This adds explicit handling for reusable objects such as retrieved context, prefix cache entries, semantic cache entries, and selected tool outputs.
- **Option C: trajectory/DAG memory fabric.** This exposes branch state, verifier state, trajectory logs, tool-output replay, and durable workspace state for agentic workflows with branch-and-merge execution.

The earlier synthesis stated that controls should remain Option A; retrieval-augmented generation (RAG) can justify Option B only when object reuse is safe and useful; and code-agent, verification-heavy, or multi-agent branch/merge workloads can justify Option C only when trajectory, verifier, branch, and durable-state value survive overheads and correctness gates.

Cycles 16-18 sharpened that conditional claim. Cycle 16 asked whether the energy and economics argument survives when per-byte energy, dollar cost, and pooled-memory contention are treated as deferred constants rather than assumed benefits. Cycle 17 asked whether security and provenance gates can be made causal in replay rather than documented as external caveats. Cycle 18 is represented by the supplied reporter range and session IDs, with no distinct additional named artifact found in the workspace.

Approach

The report uses the following source material:

Cycle	Source IDs and artifacts	Role in timeline
Cycle 16	worker 1db063f4-6b93-482e-bf52-d314cae93b44; ledger events for M-ENERGY-1; memory-centric-agentic/energy_economics_contention.md; scripts/evaluate_energy_economics.py; scripts/plot_energy_economics.py; data/energy_*.csv; data/cxl_contention_thresholds.csv; energy figures	Built and validated the energy/economics and CXL-contention falsification
Cycle 17	researcher 3939f8b4-5383-4ebd-8513-ac814fdaf622; worker 9b1d395e-0c60-4692-92c5-5d925a5d3d53; auditor e9226616-3371-4203-a2da-ea6317516ffd; supplied audit report; memory-centric-agentic/security_telemetry_enforcement.md; scripts/security_enforcement_replay.py; tests/verify_security_enforcement_replay.py; security trace-v3 tables and figures	Built, audited, repaired, and validated executable security-enforcement replay.
Cycle 18	researcher b0e96d70-84fc-4043-864f-ad06d56328f6; worker f97f3e40-19b2-45c7-b805-f3bda5cbf649; auditor d5fb087e-8a85-4a77-bced-6d17629d45ba	No separate cycle-18 artifact surface found in the workspace. The cycle is treated as part of the reporting boundary.
Context immediately before cycle 16	M-EXP-1 integration rows, tests/verify_mexpl_integration.py promise_ledger.jsonl, plan_of_record.md	Explains why cycle 16 started from validated DC-003 through DC-006 measurement designs and a repaired ledger chronology.

Evidence labels are preserved from the source artifacts:

Option A/B/C robustness under per-byte energy/cost sweeps.

Figure 1: Option A/B/C robustness under per-byte energy/cost sweeps.

CXL/pooled-memory latency thresholds where warm-tier placement reverses.

Figure 2: CXL/pooled-memory latency thresholds where warm-tier placement reverses.

- **sourced** means grounded in listed public references or accumulated calibration artifacts.
- **derived** means produced by the project’s analytical equations or rule logic.
- **simulated** means produced by deterministic synthetic sweeps or replay harnesses.
- **measurement_design** means it defines what production telemetry must collect.
- **speculative** means the package still lacks production or benchmark measurement for the claim.

Results

Cycle 16: Energy, Economics, and CXL Contention

Cycle 16 added M-ENERGY-1, a falsification harness for energy, dollar, and warm-tier contention claims. Its governing expression is:

$$NetEnergyValue = E_{recompute_avoided} + E_{movement_avoided} - E_{residency} - E_{transfer} - E_{validation} - E_{coordination}.$$

The harness treats this as a sensitivity model, not a measured deployment result. It consumes prior calibration, cost-model, queueing, runtime, synthesis, and measurement-threshold artifacts, then emits synthetic rows showing when memory-centric choices survive or collapse.

The cycle focused on two deferred constants:

- **DC-001:** per-tier energy and dollar cost per byte moved or retained.
- **DC-002:** CXL or pooled-memory p50, p95, and p99 latency under contention.

The key decision from M-ENERGY-1 is that the economic claim CL-012 remains speculative. The harness can define thresholds and measurement requirements, but it cannot convert energy or dollar value into a calibrated claim without production telemetry.

The architecture-sensitivity table has 96 rows. Across the synthetic sweep, the resulting option distribution was:

Option after sweep	Rows
Option A	48
Option B	40
Option C	8

The collapse reasons were:

Collapse reason	Rows
No synthetic collapse under this sweep	47
DC-002 tail contention exceeds retained-value margin	25
DC-001 zero removes energy/dollar evidence; retained-value-only claim is not economic support	12
Synthetic net value changes option after energy/contention charges	12

This result preserves the earlier architecture boundary. Controls remain Option A. Option B and Option C require positive retained value before energy or dollar credit is counted. If DC-001 is zero, the energy/economics argument disappears even if latency, capacity, or correctness arguments remain. If DC-002 tail latency exceeds the retained-value margin, warm-tier placement is downgraded.

Synthesis claims updated by DC-001/DC-002 measurement outcomes.

Figure 3: Synthesis claims updated by DC-001/DC-002 measurement outcomes.

The CXL threshold table has 48 rows: 23 rows where the synthetic warm tier still helps and 25 rows where it should be downgraded. For example, the RAG Option B object group had a benefit margin and collapse threshold of 10.8 synthetic units, so p50, p95, and p99 settings at 0.5, 2.0, and 8.0 remained below the threshold, while a pathological p99 setting at 1000.0 forced a downgrade. For an Option C object group in the same workload, the collapse threshold was lower, 5.5, so an 8.0 p99 setting already triggered downgrade.

The claim-update table keeps CL-012 speculative and marks other claims as calibration-ready rather than proven by the synthetic sweep. The required production telemetry is explicit:

Measurement ID	Quantity	Purpose
DC001-BYTE-ENERGY-001	Per-tier joules per byte moved and retained	Determines whether measured byte movement or residency savings exceed validation and coordination energy.
DC001-DOLLAR-001	Dollar cost per retained byte and moved byte	Determines whether retained state reduces cost after tier occupancy, transfer, and capacity reservation are included.
DC002-CXL-CONTENTION-001	CXL or pooled-memory p50/p95/p99 latency under contention	Determines whether warm-tier latency tails exceed retained-value margins.
SECURITY-GATE-ENERGY-001	Authorized safe reuse rate before energy credit	Prevents unsafe, stale, or unauthorized reuse from being counted as energy savings.

The cycle 16 auditor validated the harness without code changes. The ledger records 288 scenario rows, 96 architecture-sensitivity rows, 48 CXL threshold rows, 7 claim-update rows, 4 measurement rows, nonblank figures, successful plot regeneration, successful Python compilation, and green `promise_check/org_check`.

Cycle 17: Security Telemetry and Enforcement Replay

Cycle 17 added M-SECOPS-1, which turns the earlier security/provenance model into executable replay. The replay rule is:

$$SecurityAdjustedValue(object) = RawRetainedValue - ValidationOverhead - ExpectedSecurityLoss.$$

Raw retained value is credited only when telemetry proves authorized reuse. If a gate fails, the replay assigns zero safe reuse credit and chooses deny, downgrade, recompute, or architecture fallback.

Trace-v3 extends `data/agentric_trace_events_v2.csv` with fields needed for security-grade reuse decisions:

- `tenant_scope`
- `cache_salt`
- `actor_id`
- `replay_authorization_scope`
- `verifier_evidence_hash`
- `retention_hold_state`
- `pointer_valid`
- `validation_gate_ids`
- `validation_decision`
- `validation_lookup_count`
- `validation_queue_wait`
- `validation_start_time`

Raw reuse credit is split into safe, denied, downgraded, and overhead-dominated retained-value credit by workload.

Figure 4: Raw reuse credit is split into safe, denied, downgraded, and overhead-dominated retained-value credit by workload.

Validation-gate p95, p99, and max synthetic latency units are shown for each gate; tails are largest for trajectory, retention, verifier, and pointer checks.

Figure 5: Validation-gate p95, p99, and max synthetic latency units are shown for each gate; tails are largest for trajectory, retention, verifier, and pointer checks.

- `validation_end_time`

The represented gates are provenance presence, source freshness, tenant isolation, cache-salt isolation, trajectory lineage, replay authorization, verifier evidence binding, retention/hold compliance, and pointer recoverability.

The generated enforcement table has 268 replay decision rows:

Decision	Count
safe reuse	123
denied reuse	75
downgraded reuse	48
overhead-dominated reuse	4
not reuse candidate	18

The fixture table has 13 rows: 2 valid controls and 11 invalid fixtures. The invalid fixtures cover missing provenance, stale source version, invalidation signal, tenant mismatch, cache-salt mismatch, unauthorized actor, contaminated lineage, tampered verifier hash, expired retention without hold, invalid pointer, and missing validation timing. After the audit patch, fixture outcomes are computed through the same `decision_for` path as replay rows.

The replay recomputes architecture decisions from safe reuse credit and gate overhead rather than annotating prior choices. The resulting architecture updates were:

Workload	Before	After security	Safe hit rate
RAG	B	B	0.740741
batch summarization/offline inference control	A	A	0.0
code-agent loop	C	C	0.53125
multi-agent branch/merge	C	C	0.40404
single-turn chat control	A	A	0.0
verification-heavy	C	A	0.483333

The most visible change is that verification-heavy collapses from Option C to Option A after security enforcement. RAG remains Option B, and code-agent and multi-agent branch/merge workloads remain Option C under the synthetic assumptions. Controls remain Option A.

The field-ablation replay has 54 rows, with 32 causal rows where removing a security field changes safe credit or the recomputed option. This makes fields such as tenant scope, cache salt, actor authorization, provenance pointers, source version, lineage, verifier evidence, retention state, and pointer validity causal in the replay rather than decorative telemetry.

The supplied audit report identified and fixed one moderate enforcement-validity defect. Before the fix, invalid fixtures reported actual outcomes without running through the same decision function used for replay rows, and several semantic checks were not enforced. The auditor patched `scripts/security_enforcement_replay.py` so fixtures execute through `decision_for`, added tenant/cache-salt matching, actor authorization, hard deny on missing provenance, and missing validation timing checks, then updated `memory-centric-agentic/security_telemetry_enforcement.md`.

Post-fix checks passed:

Architecture options before and after enforcement show controls staying Option A and verification-heavy collapsing from Option C to Option A under synthetic enforcement loss.

Figure 6: Architecture options before and after enforcement show controls staying Option A and verification-heavy collapsing from Option C to Option A under synthetic enforcement loss.

- `python3 scripts/security_enforcement_replay.py`
- `python3 scripts/plot_security_enforcement.py`
- `python3 tests/verify_security_enforcement_replay.py`
- Python compilation
- `promise_check`
- `org_check`

The audit decision was **VALIDATED**.

Cycle 18: Reporting Boundary and Artifact Surface

The supplied cycle range includes cycle 18 with researcher, worker, and auditor session IDs. In the workspace, no separate cycle-18 named milestone, markdown artifact, script, or ledger entry was found beyond the validated cycle 16 and 17 artifacts and this reporter consolidation boundary. The report therefore treats cycle 18 as a consolidation/reporting interval rather than a new technical milestone.

This is a record gap, not a technical defect in the artifacts. The supplied IDs remain traceability anchors:

- Cycle 18 researcher: `b0e96d70-84fc-4043-864f-ad06d56323f6`
- Cycle 18 worker: `f97f3e40-19b2-45c7-b805-f3bda5cbfe6e`
- Cycle 18 auditor: `d5fb087e-8a85-4a77-bced-6d17629d45ba`

Discussion

Cycles 16-18 changed the package in three concrete ways.

First, energy and economics are now bounded by falsification conditions. The project no longer needs to say only that memory movement might reduce energy or cost. It now has telemetry requirements and synthetic collapse thresholds for per-tier byte energy, per-byte cost, capacity reservation, CXL latency tails, and safe reuse. The conclusion is conservative: if measured DC-001 values are zero or below noise, energy is not a support argument; if DC-002 p95 or p99 contention exceeds retained-value margins, warm-tier placement is downgraded.

Second, security and provenance moved from risk analysis to executable enforcement. Earlier artifacts stated that unsafe reuse should not count. M-SECOPS-1 implements that principle in replay. Missing provenance, tenant mismatch, cache-salt mismatch, unauthorized actor, verifier tampering, expired retention, invalid pointers, and missing validation timing all deny or downgrade reuse. The architecture decision is recomputed from safe credit, not raw reuse.

Third, the package's main thesis became narrower but more defensible. The current version is:

MemoryCentricArchitecture is strongest when $RetainedStateValue > CoordinationCost + ValidationOverhead + RecoveryCost$

The thesis is not that every agentic workload should use a memory-centric architecture. It is that some workloads, especially RAG and branch/merge agentic systems, can justify explicit memory-object or trajectory-state management only when the relevant state passes safety, provenance, latency, and economics gates.

Conclusions and Recommendations

The cycles 16-18 package is validated as a measurement-ready extension of the earlier synthesis.

The immediate architecture position is:

1. Keep Option A as the default for controls and workloads without positive safe non-KV retained value.
2. Use Option B for RAG-like object reuse only when provenance, freshness, isolation, invalidation, safe-hit rate, validation overhead, and DC-001/DC-002 economics survive measurement.
3. Use Option C for agentic branch, verifier, trajectory, tool-output replay, and durable workspace state only when replay authorization, verifier integrity, retention compliance, durable tails, queueing overhead, and CXL/contention costs remain below retained-value margins.

Recommended next work:

1. Measure DC-001 and DC-002 on target hardware and workloads. The energy/economics claim remains speculative until per-tier energy, cost, residency, transfer, and pooled-memory contention telemetry replace synthetic proxy rows.
2. Convert trace-v3 enforcement replay into a runtime or benchmark harness. The current replay is synthetic but now has executable gates, fixtures, ablations, and architecture recomputation.
3. Prioritize verification-heavy workloads as a falsification target. They are the clearest case where raw Option C value can collapse to Option A after security enforcement.
4. Preserve the rule that unsafe reuse contributes zero positive value before any energy, dollar, latency, or capacity benefit is counted.

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Appendix: Implementation Details

Code Organization

Cycle 16 added or updated the following primary artifacts:

Artifact	Purpose
memory-centric-agentic/energy_economics_contention.py	Provide summary of the DC-001/DC-002 energy, dollar, and contention harness.
scripts/evaluate_energy_economics.py	Generates scenario, sensitivity, CXL threshold, claim-update, and measurement-requirement CSVs.
scripts/plot_energy_economics.py	Renders energy sensitivity, CXL threshold, and claim-update figures.
data/energy_economics_scenarios.csv	288 synthetic scenario rows.
data/energy_architecture_sensitivity.csv	96 architecture-sensitivity rows.
data/cxl_contention_thresholds.csv	48 CXL/pooled-memory threshold rows.
data/energy_claim_update_matrix.csv	7 claim-update rows.
data/energy_measurement_requirements.csv	4 production telemetry requirement rows.

Cycle 17 added or updated the following primary artifacts:

Artifact	Purpose
memory-centric-agentic/security_telemetry_enforcement.py	Provide summary of trace-v3 fields, enforcement results, architecture updates, and limits.
scripts/security_enforcement_replay.py	Generates trace-v3 security rows, enforcement decisions, fixtures, ablations, and architecture updates.
scripts/plot_security_enforcement.py	Renders safe-reuse waterfall, gate-latency, and option-update figures.
tests/verify_security_enforcement_replay.py	Verifies trace-v3 fields, gates, invalid fixtures, safe credit, architecture changes, ablations, and figures.
data/security_trace_v3_events.csv	503 trace-v3 event rows.
data/security_enforcement_decisions.csv	268 replay decision rows.
data/security_invalid_trace_v3_fixtures.csv	13 fixture rows.
data/security_field_ablation_results.csv	54 ablation rows.
data/security_architecture_decision_updates.csv	6 architecture-update rows.

Context carried into cycle 16 from the prior integration repair:

Artifact	Purpose
tests/verify_mexpl_integration.py	Verifies parent M-EXP-1 integration across DC-003, DC-004, DC-005, and DC-006.
data/measurement_experiment_specs.csv	25 rows: DC-005 7, DC-006 7, DC-004 6, DC-003 5.
data/measurement_required_fields.csv	30 rows: DC-005 13, DC-006 8, DC-003 5, DC-004 4.
data/measurement_thresholds.csv	18 rows: DC-004 5, DC-003 5, DC-005 4, DC-006 4.
data/measurement_claim_update_matrix.csv	29 rows: DC-005 11, DC-006 7, DC-003 6, DC-004 5.
data/measurement_synthetic_probe_results.csv	17 rows: DC-005 6, DC-006 5, DC-004 3, DC-003 3.

Figure Inventory

The following figures were checked for dimensions during reporting:

Figure	Dimensions
data/energy_architecture_sensitivity.png	1920 x 960
data/cxl_contention_thresholds.png	2240 x 960
data/energy_claim_update_map.png	1760 x 800
data/security_safe_reuse_waterfall.png	1920 x 960
data/security_gate_latency_distribution.png	1920 x 960
data/security_option_update_matrix.png	1280 x 960

Validation Results

Validation commands run or reported during this reporter pass:

Command	Result
<code>python3 tests/verify_mexpi_integration.py</code>	PASS; 19 checks covering DC-003 through DC-006 integration and placeholder absence.
<code>python3 tests/verify_security_enforcement_replay.py</code>	PASS; 503 trace rows, 268 decision rows, 11 invalid fixtures, 9 represented gates, 1 security option change.
<code>python3 -m long_exposure.tools.org_check <workspace></code>	PASS.
<code>python3 -m long_exposure.tools.promise_check .</code>	PASS; 92 events and 16 plan milestones.

Validation reported by the supplied audit report for M-SECOPS-1:

Check	Result
<code>python3 scripts/security_enforcement_replay.py</code>	PASS.
<code>python3 scripts/plot_security_enforcement.py</code>	PASS.
<code>python3 tests/verify_security_enforcement_replay.py</code>	PASS.
Python compilation	PASS.
<code>promise_check</code>	PASS; 92 events and 16 plan milestones.
<code>org_check</code>	PASS.
Security figures	Regenerated and nonblank.

Source Session References

Cycle	Role	Session ID
16	worker	1db063f4-6b93-482e-bf52-d314cae93b44
17	researcher	3939f8b4-5383-4ebd-8513-ac814fdaf622
17	worker	9b1d395e-0c60-4692-92c5-5d925a5d3d53
17	auditor	e9226616-3371-4203-a2da-ea6317516ffd
18	researcher	b0e96d70-84fc-4043-864f-ad06d56323f6
18	worker	f97f3e40-19b2-45c7-b805-f3bda5cbfe6e
18	auditor	d5fb087e-8a85-4a77-bced-6d17629d45ba

No callable session-search or catalog tool was available in this environment. The report therefore uses the supplied session IDs for traceability and bases technical claims on workspace artifacts, `promise_ledger.jsonl`, `plan_of_record.md`, validation command outputs, and the supplied audit report.

Cross-Reference Map

Origin	Consuming artifact	Flow
<code>data/calibration_deferred_constants.csv</code>	<code>scripts/evaluate_energy_economics.py</code>	Defines DC-001 and DC-002 as unresolved constants for sensitivity sweeps.
<code>data/cost_model_scenarios.csv</code>	<code>scripts/evaluate_energy_economics.py</code>	Supplies energy and dollar proxy inputs.
<code>data/queueing_architecture_winners.csv</code>	<code>scripts/evaluate_energy_economics.py</code>	Supplies queueing-derived architecture context.
<code>data/runtime_workload_summary.csv</code>	<code>scripts/evaluate_energy_economics.py</code>	Supplies retained-size and net-value proxies.

Origin	Consuming artifact	Flow
data/synthesis_architecture_decisions/inputs/evaluate_energy_economics.py	scripts/security_enforcement_replay.py	Provides baseline Option A/B/C decisions for sensitivity and enforcement updates.
data/agentive_trace_events_v2.csv	scripts/security_enforcement_replay.py	Provides synthetic trace events extended into trace-v3 security telemetry.
data/measurement_required_fields.csv	scripts/security_enforcement_replay.py	Supplies DC-006 security/provenance fields used as enforcement inputs.
data/security_enforcement_decisions/inputs/evaluate_security_architecture_decisions.py	data/synthesis_architecture_decisions/inputs/evaluate_energy_economics.py	Safe updates and validation overhead recompute architecture options.
tests/verify_security_enforcement_replay.py	data/synthesis_architecture_decisions/inputs/evaluate_energy_economics.py and security figures	Verifies enforcement behavior, ablations, option changes, and figure validity.
MANIFEST.md	Future researcher and worker turns	Updated snapshot of scripts, artifacts, stats, cross-references, and validation state after this report.